Noel Burton-Krahn

Scientific Software Engineer / Consultant

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Available in the EU and Canada

PROFILE

Scientific software engineer with more than 30 years of experience building algorithms and reliable software for biomedical devices, scientific instruments, autonomous systems, and distributed infrastructure. Hands-on C/C++ and Python expertise spans ECG/EEG signal processing, machine learning, Linux, and real-time systems, from research prototypes through regulated products.

TECHNICAL FOCUS

Scientific and biomedical: algorithms, signal processing, ECG/EEG, physiological data, machine learning/deep learning (PyTorch, U-Net), numerical and research software

Robotics and instrumentation: PX4 autonomy, real-time control, hardware integration, radio-astronomy correlator simulation, remote sensing

Systems: Linux, distributed systems, fault tolerance, networking, OpenStack/SDN, cloud infrastructure, data systems

Languages and tools: C/C++, Python, Matlab/Octave, Perl, C#/.NET, Java, SQL, PyTorch, Git, Make/CMake, Docker, Kubernetes, Google Cloud, AWS, Cursor

RECENT SCIENTIFIC & TECHNICAL EXPERIENCE

ECG Algorithm Development Contractor · HeartEye — May 2026 – Oct 2026

Developed a PQRST detection algorithm in Python and C++ for a 12-lead embedded ECG device, combining conventional signal processing with a PyTorch U-Net model.

Software Team Lead · Avy Drone Networks — Nov 2024 – Aug 2025

Led and mentored the software team developing an autonomous fixed-wing aircraft, coordinating software projects and working closely with hardware engineers. Supervised four developers and hired two more.

Contributed as a senior developer to the PX4-based system using C++ and Qt in a real-time, Linux-based robotics environment.

Head of Software · Prolira — May 2017 – Aug 2024

Lead software architect and developer; managed engineering, quality and regulatory processes, and technical documentation. Developed software for a medical device that received FDA approval.

Built a substantial EEG analysis and AI platform in C++ and Python, combining digital signal processing and machine learning.

Developed algorithms in MATLAB and ported MATLAB code to production C++.

Built Google Cloud data and ML-training infrastructure and a portal for remote device acquisition and customer support.

Data Scientist · Clinezeman — Apr 2017 – Jan 2019

Developed a machine-learning application for biomedical signal processing using C++, Python, Matlab/Octave, mathematical algorithms, and neural-network methods; this work continued into Prolira.

Senior Distributed Systems Engineer · Cisco / Piston — Jun 2013 – Apr 2017

Developed Linux and Python system-control software for OpenStack clusters, software-defined networking, and virtualization.

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SELECTED SCIENTIFIC AND SYSTEMS WORK

Lead Engineer · Gatekeeper Systems — Apr 2003 – Feb 2006

Delivered a portable Linux digital-video surveillance system and Windows playback software using C/C++, Perl, OpenGL, VB.NET, and SQLite.

Research Developer · CardioComm / Harley Street Software — May 1994 – Jun 1998

Developed Windows ECG applications and signal-processing software, including the P-QRS-T feature-detection algorithm covered by US Patent 5,758,654; C/C++, Visual Basic, MFC, OLE, and ActiveX.

SELECTED ALGORITHM AND SYSTEMS PROJECTS

Research Programmer · Dominion Radio Astrophysical Observatory — Jan 1992 – May 1992

Designed and developed a Matlab and C simulation of a hardware correlator combining radio-antenna signals on Sun SPARC and IBM RISC 6000 workstations.

Selected Independent Projects

HotSwap · Transparent Linux Server Failover — Published 2002

Developed a Linux kernel/server-replication system for seamless failover, preserving active TCP connections, internal memory state, and external network state. Presented at USENIX LISA XVI and covered by a US patent application.

BKBox and NatNix · Linux and Networking Products

Built a Linux office server providing OpenAFS remote files, secure IMAP, web services, and firewall/gateway functions; developed a C++, firewall-tunnelling peer-to-peer VPN for Linux and Windows.

EARLIER EXPERIENCE

2012–2013: Kal Tire — point-of-sale and inventory systems; C#/.NET, Groovy, SQL.

2008–2012: Clover Point, Telocate, Eightfold Logic — GIS, remote-device tracking, and analytics; Python, C#, Oracle, PostgreSQL, Perl.

2006–2008: Washington State DOC and Kodak — Java/WebSphere/DB2 mainframe integration and commercial-printing software.

1998–2003: Kronofusion, Mercurial, FaxPC, BC Government — encrypted TCP tunnelling, remote desktop, Linux/Windows applications, and technical leadership.

Co-op Software Engineer / Research Programmer · Power Measurement / Siemens; Pacific Forestry Centre — Sep 1992 – Sep 1993

Developed real-time C++ power-meter communications and a Prolog expert system integrating remote sensing, image processing, GIS, and modelling.

SELECTED PUBLICATIONS AND PATENTS

ECG P-QRS-T Onset, Offset and Peak Detection Method and Apparatus. US Patent 5,758,654, Harley Street Software, 1998.

HotSwap — Transparent Failover for Linux Servers. USENIX LISA XVI, 2002.

Method and apparatus for providing transparent fault tolerance within an application server environment. US Patent Application 2004/0153709, 2004.

EDUCATION

M.Sc., Computer Science, University of Victoria, 2003.
B.Sc. (Honours), Computer Science / Mathematics / Physics, University of Victoria, 1994

REFERENCES

Available on request